Delivery Route Graph CRUD – Package Delivery Service

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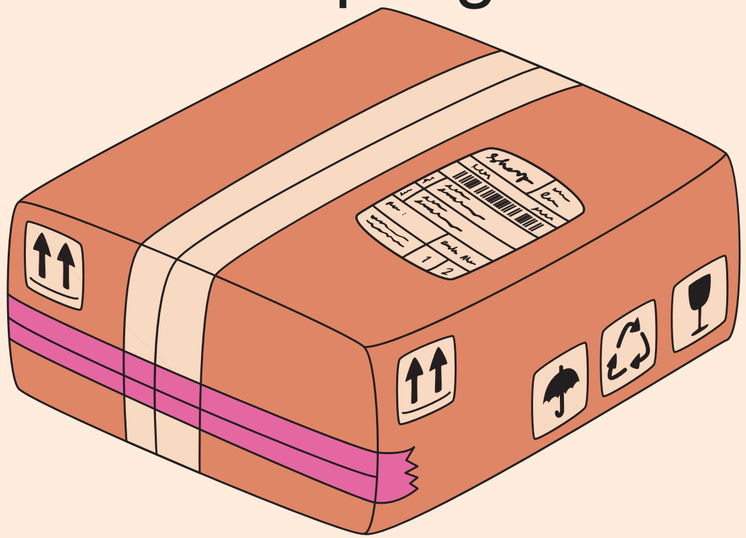
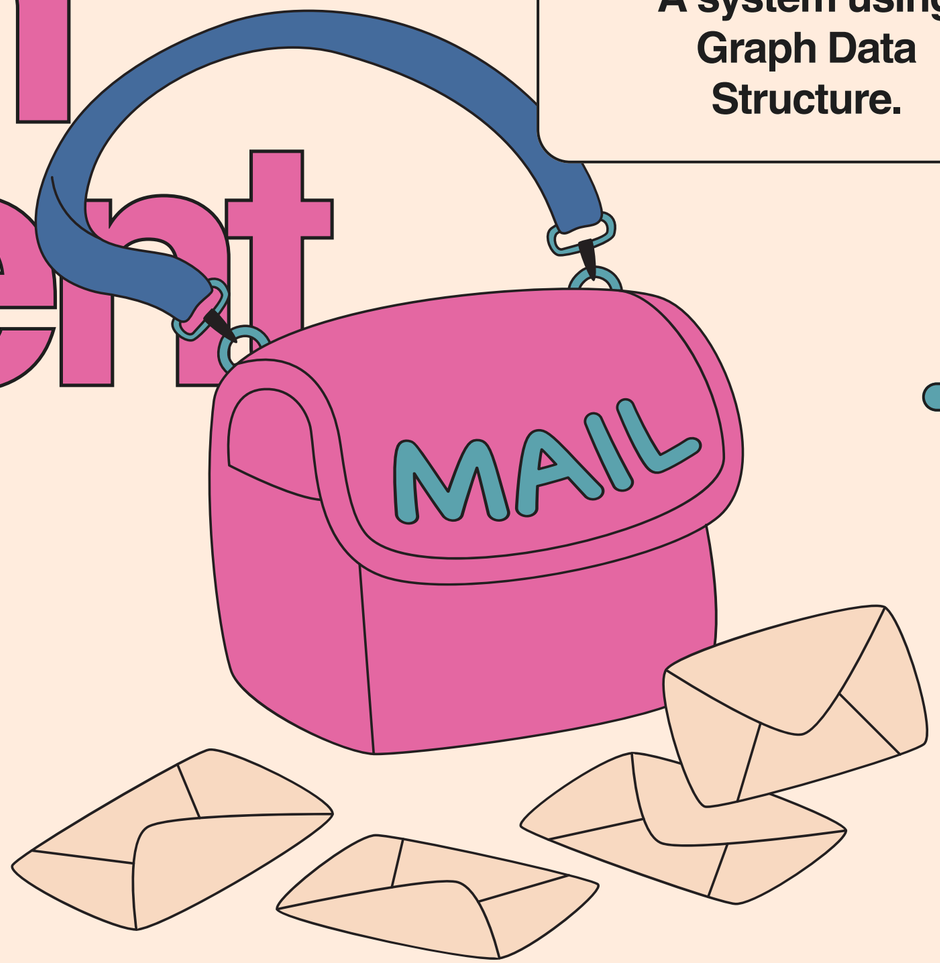
Introduction

- Package delivery systems manage routes between locations.
- Efficient route management is important for logistics.
- Manual handling is difficult and error-prone.
- Data structures help organize and manage routes.



Problem Statement

A system using Graph Data Structure.



Delivery companies need a system to:

- Add delivery routes
- Update route distances
- Delete routes
- Search routes quickly

Objective

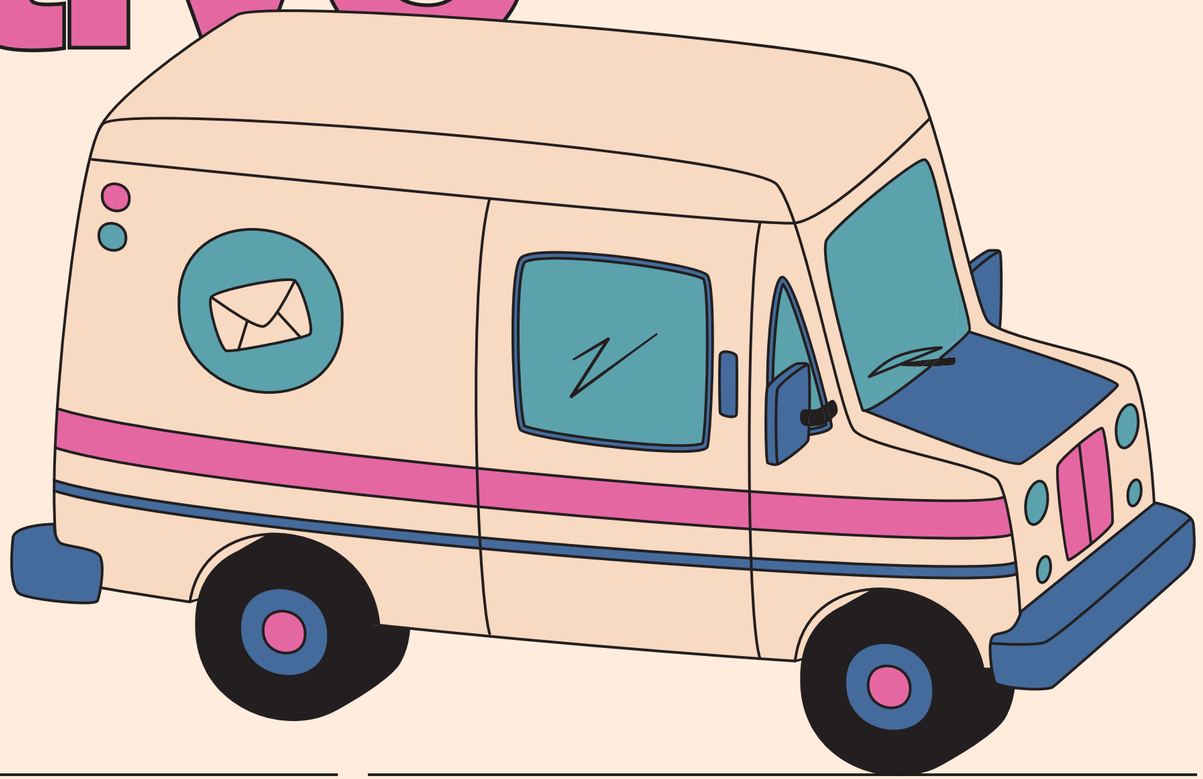
Manage delivery routes efficiently

Implement real-world application in C

Perform CRUD operations

Use Graph Data Structure

Create a menu-driven program



Data Structure Used

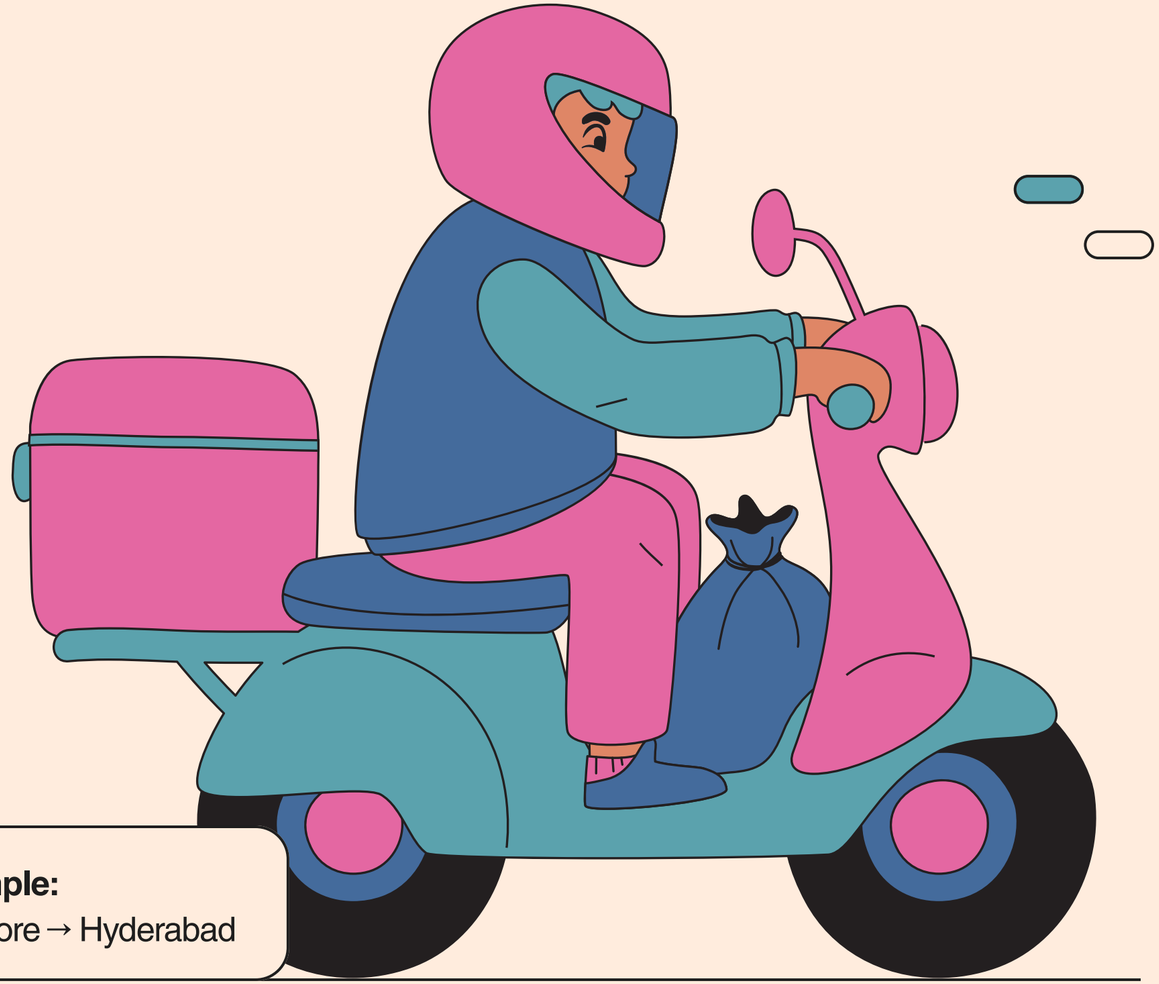
Graph Data Structure

Nodes → Locations

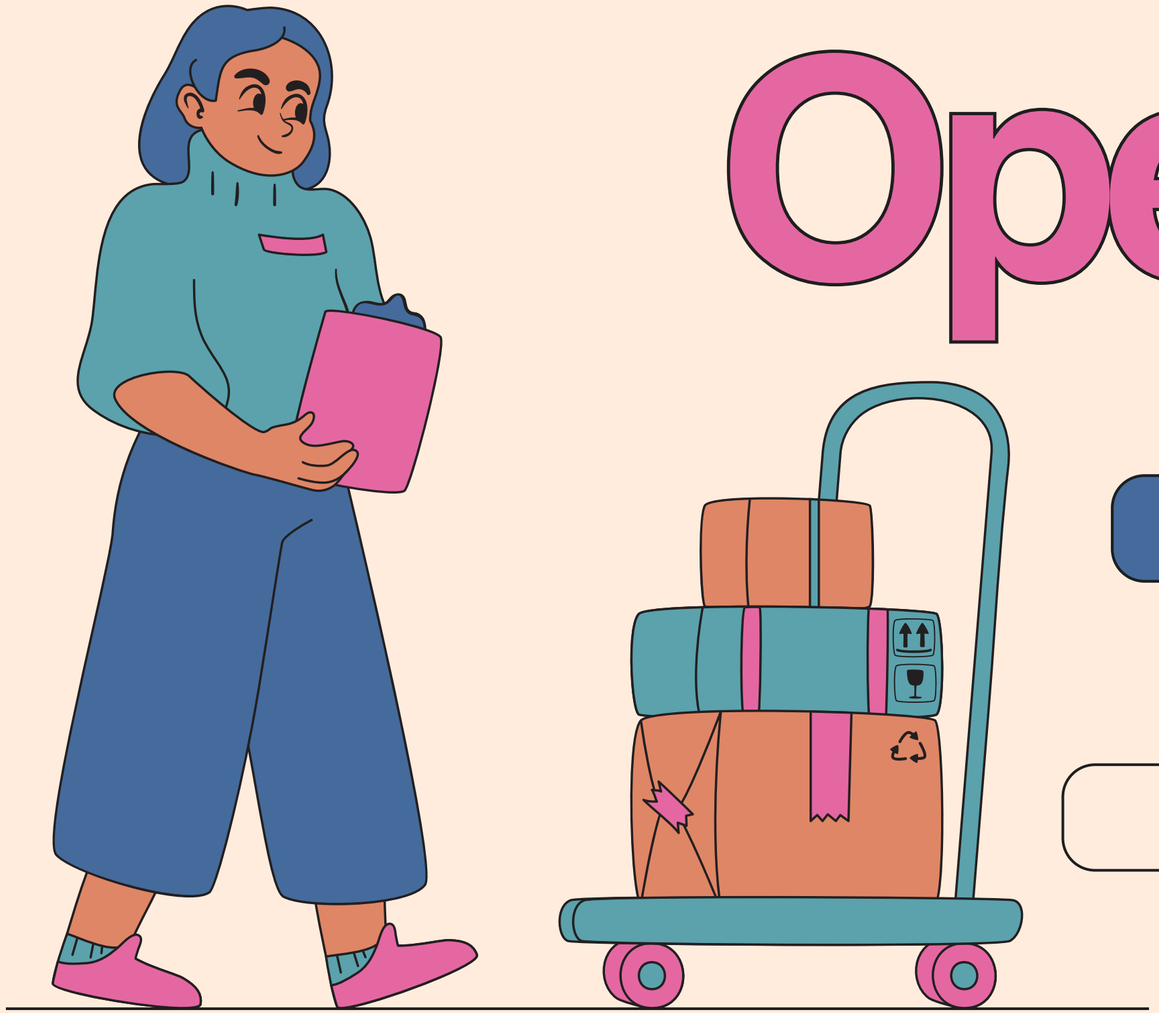
Edges → Routes

Weight → Distance

Example:
Chennai → Bangalore → Hyderabad



CRUD Operations



Create
Add location and route

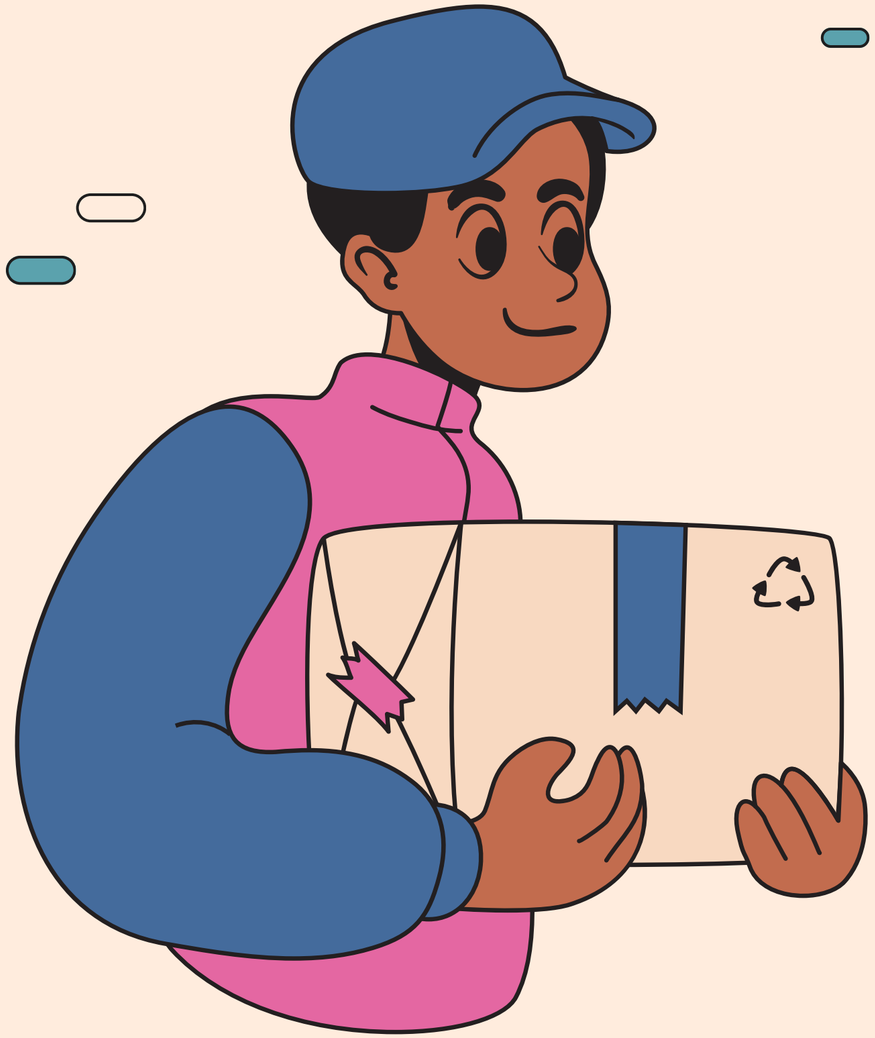
Read
Display routes

Delete
Remove routes

Update
Modify distance

Program Features

- Add locations
- Add routes
- Update routes
- Delete routes
- Search routes
- Display all routes



GitHub Repository

Project structure:

src → main.c
docs → project report
ppt → presentation
README.md
sample_output.txt
video_demo_link.txt

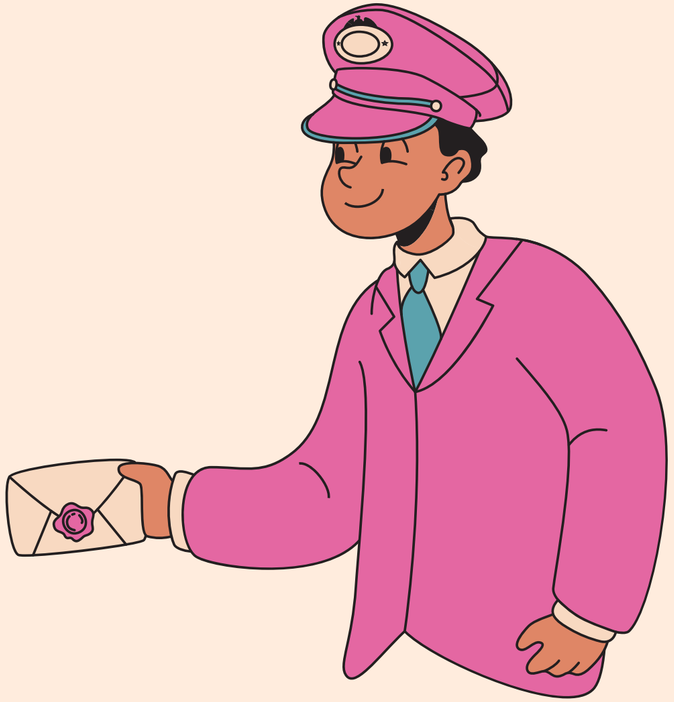
Both team members contributed using commits.

Sample Output

Add Location: Chennai
Add Location: Bangalore

Add Route:
Chennai → Bangalore
Distance: 350 km

Display:
Chennai → Bangalore : 350 km



Conclusion

Implemented graph-based delivery system

Used C programming

Performed CRUD operations

Demonstrated real-world application

Thank You